# SARAVANA KUMAR B

## AI/ML ENGINEER

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Ooty

## Profile Summary

Results-driven AI Engineer with 4+ years of experience at TCS, skilled in AI model development, NLP, and computer vision. Proficient in Python and Machine Learning. Proven problem-solver, collaborative team player, and dedicated to staying current with AI trends. Seeking an impactful role to drive innovation, growth, and success in cutting-edge AI solutions.

## Skills

- Python, FastAPI
- Data Science, statistics
- Azure, AWS, GCP
- MySQL DB
- scikit-learn
- numpy,pandas
- Pytorch,Tensorlfow, keras
- OpenCV, pinecone
- NLP, nltk, spacy
- Langchain, Langgraph
- LlamaIndex, RAG, MLflow
- seaborn, plotly, matplotlib

## Education

X Grade

| | | |
|---|---|---|
| Mountain Home School & Junior College | ICSE \| 74% | 2014 - 2015 |

XII Grade

| | | |
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| Stanes Anglo-Indian Hr.Sec School | State Board \| 95% | 2015 - 2017 |

Bachelors in Mechanical Engineering

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|---|---|---|
| Bannari Amman Institute of Technology | CGPA - 9.3/10 | 2017 - 2021 |

## Work Experience

### System Engineer

**Tata Consultancy Services** 2021 - 2025

At TCS, I've been involved in multiple roles and projects,

1. As a Developer in Building a Pothole Detection Model that alerts vehicle drivers to avoid accidents, by object detection using HAAR Cascade algorithm and OpenCV.
2. As a content creator, creating e-learning content using the Adobe Captivate tool to make employees understand the workings of an application with animations and special effects such as lip sinking to audio for an image.
3. As a DataLake Support Engineer for Azure built applications, monitored the data flow, and created various Azure services using Python to extract data from various up-streams, transform, and send data to various down-streams.
4. 1) **Chat Assistant**: As a ML Engineer in building an intelligent assistant to solve user incidents based on historical incident data. A new method called RAG(Retrieval Augmented Generation) was used. It helped users to quickly solve alerts without having to contact several teams. Open-source LLMs were used which enabled the organization to maintain data privacy and integrity. Tools Used: Sentence Transformers, Opensource LLM(Mistral), flask

2) **SNow KB Aritcle Tagging**: ServiceNow is a major incident management tool across many of our clients, and automation of mundane tasks greatly increases efficiency and conserves company costs, I've developed an automation algorithm which uses sentence transformer models to make a similarity match for providing most accurate Knowledge articles for oncoming incidents so that users can solve it themselves before raising an incident and auto resolve it if satisfied.
Tools used: Pandas, Spacy, Sentence Transformers, Django

3) **Data Trends Prediction** : There are multiple applications which streams data to other applications using numerous servers with huge storage capacities, these data tend to vary in size based on the business need and it was tedious for the resource allocators to be prepared for the oncoming data size and quantity to be able to properly allocate storage and GPUs. So to help them, I've developed a ML model with great accuracy which predicts the data count for all the apps for each day and showed it as a data trends dashboard so that the users can view them and anticipate it and be ready with the resources accordingly.
Toosl used: StatsModel, Pandas, Arima Interpolation, Sci-kit learn

4) I**ntelligent Change Request Management** : ServiceNow change requests are generated in thousands every month in any organization and they have to be carefully reviewed by a change manager before its approved by the leads of the organization, this review process is a bit tedious and time consuming also changes are prone to failure and cause business disruptions, in order to make the review process easier for the chnage reviewers we have developed an AI model using transformer architecture by utilising the past years change data we assist the reviewers by fully analysing and reviewing the whole change field by field and predict the success rate of the change which made the whole review process seamless and saved upto $1 million in revenue.
Toosl used: transformers, Random Forest, tensorflow, Sci-kit learn

5) **Incident Resolution Recommender**: Throughout the years several incidents have been resolved in ServiceNow where resolution notes have been created by the resolvers. These notes are crirical in solving recurring issues so there is always a team dedicated to show these resolution while incidents occurs but since its a manual effort we developed an incident recommender that recommends the right resolution for a recurring incident. The app uses semantic embeddings to store the resolution notes and performs a similarity search with the incoming query to find the most appropriate resolution with contextual understanding which was gained while training. This greatly resolved recurring incidents in blazing fast speed which again saved a huge cost for the company.
Tools Used: sentence-transformers, RAG.

6) **Dashboard Metrics Chat Assistant**: A client needed an assistant for a PowerBi Dashboard to get metrics in a natural language like interaction with an assistant and get a summary of the metrics. I've developed the bot by using Azure Cognitive services by using OpenAI's ChatGPT api to provide the context to the model and get the results based on the users query.
Tools Used: requests, Django, api integration.

## Self-Delevoped Projects

### 1. Pothole and Animal detection on the Road to prevent Accidents

It has been a very serious issue that vehicles get into accidents or collapse due to huge or small potholes on the road, and animals suddenly crossing the road to puzzle drivers leading to serious injuries. In order to prevent this scenario a model was developed to detect potholes and animals ahead calculating their distance and potential impact with the vehicle and alerting the driver to be cautious. Thanks to powerful and state of the art YOLO algorithm.
Tools Used: Pyhton, Tensorflow Keras, CNNs, open-cv YOLO.

### 2. Using open-source LLMs to query private documents locally

With the revolution of Large Language Models, it's tempting to prompt the company to confidential data to solve complex production problems and to make the work faster, which might lead to data leaks and is a serious concern among companies. In order to avoid this scenario, a software was developed to query confidential documents locally using OpenAI's chatGPT via API key or GPT like performing open-source LLMs like LLAMA and Mistral without exposing confidential company information to others. Thanks to the open-source LLM contributors and RAG.
Tools Used: Python, NLP, Langchain, Embeddings, Vector bases, Streamlit, llamaIndex

### 3. An AI chess Engine to mimic the legendary Mikhail Tal's playing style

Being a chess enthusiast I've always wanted to play gambits and tactical chess unlike positional and slow games. Mikhail Tal was a famous chess player who kept this style forefront and inspired many other chess grand masters. To mimic his style a model was developed by feeding all of Tal's games to make the model adapt to his playing style. The model makes insane sacrifices and gambits to checkmate its opponents. Thanks to Reinforcement Learning.
Tools Used: Python, pygame, chess.pgn, tensorflow

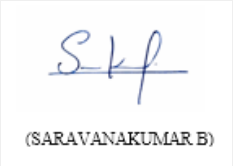
## Key Achievements

1. Provided a significant safety feature to autonomous vehicles to avoid potholes and sudden animal intrusion on road.
2. Developed 5+ hours of online trainings to the client for a huge application which made new employees to get familiar with ease.
3. Saved thousands of dollars in revenue and time by providing our client with the AI tool(SNowKB article tagging)
4. Provided a fully functional chatbot that uses RAG to help client solve ServiceNow and BigPanda tickets, created user interactive dashboards, built multiple prediction models.

## Certifications

- NLP by Deeplearning.ai

Hereby, I declare that all the above given details are true to my conscience and are verifiable.



(SARAVANAKUMAR B)